Number theory – 2

Orders and primitive roots

Homework

14 сентября 2026 года

Theoretical minimum. For $\gcd(a, n) = 1$ the order $\operatorname{ord}_n(a)$ is the least $d > 0$ with $a^d \equiv 1$; then $a^m \equiv 1$ if and only if the order divides m, and the order divides $\varphi(n)$. Hence every prime divisor of $a^n - 1$ either divides some $a^d - 1$ with $d \mid n$, $d < n$, or is $\equiv 1 \pmod n$. The group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic, because $x^d = 1$ has at most d roots over the field $\mathbb{F}_p$ and $\sum_{d|p-1} \varphi(d) = p - 1$; a primitive root modulo n exists exactly for $n = 1, 2, 4, p^k, 2p^k$, while $(\mathbb{Z}/2^k\mathbb{Z})^* \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$ for $k \geqslant 3$. Fixing a primitive root turns multiplication into addition, so a is an m-th power modulo p if and only if $a^{(p-1)/\gcd(m,p-1)} \equiv 1$. Zsygmondy's theorem, which may be quoted, provides a primitive prime divisor of $a^n - b^n$ with two exceptions.

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1. Find all prime numbers p for which there is a unique $a \in \{1, 2, \ldots, p\}$ with $a^3 - 3a + 1$ divisible by p. (IMC)

2. Prove that for a prime $p > 3$ the number of $k \in \{1, \ldots, p-2\}$ with $I(k+1) < I(k)$ is greater than $p/4 - 1$, where $I(k) \in \{1, \ldots, p-1\}$ is the inverse of k modulo p. (Putnam 2025)

3. Let $b_0 = 0$ and $b_{n+1} = 2b_n^2 + b_n + 1$ for $n \geqslant 0$. Prove that for each $k \geqslant 1$ the number $b_{2^{k+1}} - 2b_{2^k}$ is divisible by 2^{2k+2} but not by 2^{2k+3}. (Putnam 2025)

4. Let k be a positive integer and $n = 2^k + 1$. Prove that n is prime if and only if there exist a permutation $a_1, \ldots, a_{n-1}$ of $1, \ldots, n-1$ and integers $g_1, \ldots, g_{n-1}$ such that n divides $g_i^{a_i} - a_{i+1}$ for every i, where $a_n = a_1$. (IMO shortlist 2011, N8)

5. Determine the multiplicative order of 2 modulo a Mersenne prime, prove that for $k \geqslant 2$ a prime divisor of $2^{2^k}+1$ is $\equiv 1 \pmod{2^{k+2}}$, and deduce that the Fermat numbers are pairwise coprime; then give the resulting proof that there are infinitely many primes.

6. Let g be a primitive root modulo an odd prime p. Prove that exactly $\varphi(p-1)$ residues are primitive roots, that the product of all primitive roots is $\equiv 1 \pmod{p}$ for $p > 3$, and that the sum of all primitive roots is $\equiv \mu(p-1) \pmod{p}$.

7. Prove that for every n there are infinitely many primes p with $p \equiv 1 \pmod{n}$, by considering the prime divisors of the values of the cyclotomic polynomial $\Phi_n(x)$ at integers, and explain why this argument does not reach the residue classes other than 1.

8. Determine all integers $m \geqslant 2$ such that every n with $m/3 \leqslant n \leqslant m/2$ divides the binomial coefficient $\binom{n}{m-2n}$. (IMO shortlist 2012, N3)

9. Prove that there are infinitely many positive integers n such that the largest prime divisor of n^4+n^2+1 equals the largest prime divisor of $(n+1)^4+(n+1)^2+1$. (IMO shortlist 2013, N3)

10. Prove that for every prime $p > 100$ and every integer r there exist integers a and b such that p divides a^2+b^5-r.
(IMO shortlist 2012, N8)